Quinn Asena

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Data Scientist & Ecologist · Cary Institute of Ecosystem Studies, Forest Futures Lab, Millbrook, NY

Ecologist and data scientist focused on how ecosystems change over time and space, coupling process-based landscape simulation with CMIP6 climate scenarios on HPC systems to project forest biodiversity change, with a background spanning palaeoecology and statistical modelling.

PROFESSIONAL EXPERIENCE

Data Scientist *Cary Institute of Ecosystem Studies*
Millbrook, NY *2025 – Present*

- Automated HPC workflows for process-based landscape simulations across boreal North America under multiple CMIP6 climate scenarios.
- Large-scale data management using SQL and Apache Arrow; integration of climate, soil, and elevation data from distributed archives (NASA DAACs, STAC).

Postdoctoral Research Associate *University of Wisconsin-Madison*
Madison, WI *2022 – 2024*

- NSF-funded project developing state-space modelling methods to analyse palaeoecological records, part of the abrupt change in ecosystems project led by Jack Williams and Anthony Ives.

EDUCATION

PhD, University of Auckland *School of Environment*
Auckland, NZ *2017 – 2021*

- Explored virtual ecological methods for generating pseudoproxy data to assess statistical inferences under data uncertainty. Supervised by George Perry and Janet Wilmshurst.

MEnv. Environmental Science *University of York*
York, UK *2012 – 2016*

SELECTED PUBLICATIONS

- **Asena, Q.**, Williams, J., Stefanova, V., Johnson, J., Shuman, B., Ives, A. (2026). Statistical analyses of ecological multinomial time series to identify environmental drivers and biotic interactions. *Methods in Ecology and Evolution.* In press.
- **Asena, Q.**, Perry, G.L., Wilmshurst, J. (2026). Information loss in palaeoecological data from process and observer error. *Climate of the Past.*
- **Asena, Q.**, Perry, G.L., Wilmshurst, J. (2024). Is the past recoverable from the data? Pseudoproxy modelling of uncertainties in palaeoecological data. *The Holocene.*
- Parsons, M., **Asena, Q.**, Johnson, D., Nalau, J. (2024). A Bibliometric Review and Topic Analysis of Climate Justice Literature: Mapping Trends, Voices, and the Way Forward. *Climate Risk Management.*

SOFTWARE & OPEN SCIENCE

- **multinomialTS** — R package for state-space modelling of multinomially distributed ecological time-series data. (2024)
- **State-Space Methods for Palaeoecological Hypothesis Testing** — open workshop materials for state-space modelling of community and palaeoecological time series. (2024)
- **Authoring Collaborative Research Projects in Quarto** — resources for collaborative research workflows using Git, GitHub, and Quarto (ResBaz 2022–2024).
- **fisheR** — R package for calculating Fisher's Information on ecological time-series data. (2019)
- **Population Growth Explorer** — interactive Shiny app exploring population growth equations for undergraduate teaching. (2019)